AP CALCULUS AB

Unit 1 • Worksheet 1-2A • Answer key

Worksheet 1-2A • Limits to a Point of a Function

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to 3} \dfrac{x^2 - 9}{x^2 - 2x - 3}$ is

(A) 0

(B) $\frac{3}{2}$

(C) 3

(D) nonexistent

Solution

Answer: (B). $\frac{x+3}{x+1} \to \frac{6}{4}$.

Question 2

NO CALCULATOR

$\lim_{x \to 0} \dfrac{\sqrt{x+9} - 3}{x}$ is

(A) 0

(B) $\frac{1}{3}$

(C) $\frac{1}{6}$

(D) 3

Solution

Answer: (C). Conjugate: $\frac{1}{\sqrt{x+9}+3}$.

Question 3

NO CALCULATOR

$\lim_{x \to 0} \dfrac{\sin(5x)}{x}$ is

(A) 5

(B) 1

(C) $\frac{1}{5}$

(D) 0

Solution

Answer: (A).

Question 4

NO CALCULATOR

$\lim_{x\to 2^-} \dfrac{|x-2|}{x-2}$ is

(A) 1
(B) 0
(C) -1
(D) nonexistent

Solution
Answer: (C).

Question 5

NO CALCULATOR

$\lim_{x\to 4} \dfrac{x+1}{(x-4)^2}$ is

(A) 0
(B) $-\infty$
(C) $\frac{5}{4}$
(D) ∞

Solution
Answer: (D). $\frac{5}{0}$ with a positive denominator on both sides.

Question 6

NO CALCULATOR

$\lim_{x\to 0} x^2 \cos\dfrac{1}{x}$ is

(A) 1
(B) 0
(C) -1
(D) nonexistent

Solution
Answer: (B). Squeeze between $\pm x^2$.

Question 7 (Free response)

NO CALCULATOR

Evaluate each limit, showing your work. (a) $\lim_{x\to 5} \dfrac{x^2-3x-10}{x-5}$ (b) $\lim_{x\to 2} \dfrac{x^3-8}{x-2}$ (c) $\lim_{h\to 0} \dfrac{\frac{1}{2+h}-\frac{1}{2}}{h}$ (d) $\lim_{x\to 9} \dfrac{x-9}{\sqrt{x}-3}$ (e) $\lim_{x\to 0} \dfrac{\sin 2x}{\sin 3x}$

Solution
(a) $x+2 \to 7$. (b) $x^2+2x+4 \to 12$. (c) $\frac{-1}{2(2+h)} \to -\frac{1}{4}$. (d) $\sqrt{x}+3 \to 6$. (e) $\frac{2}{3}$.

Question 8 (Free response) NO CALCULATOR

Let $f(x) = \begin{cases} x^2 - 1, & x < 3 \\ 2x + 2, & x \geq 3. \end{cases}$ Find $\lim_{x \to 3^-} f(x)$, $\lim_{x \to 3^+} f(x)$ and $\lim_{x \to 3} f(x)$.

Solution
8, 8, 8.